

Algorithmic Trading
HW 8 - Single Stock Trade Strategy Optimization
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User Submission:

- Answer the five (5) questions below.
- Each question consists of an optimization problem and a graphical illustration.
- Answers and examples of the graphs are provided for each question
- Submit your Python Code to BB.

Trade Strategy Optimizations

Using your previously created Python Pre-Trade functions, develop Python code to solve the following:

1. Traders Dilemma
2. Minimize Cost
3. Price Improvement
4. Efficient Trading Frontier
5. Market Impact Constraint

MI Parameters

Use the following MI Parameters to answer all questions:

a1=883.58722

a2=0.35408

a3=0.755684

a4=0.826155

b1=0.963532

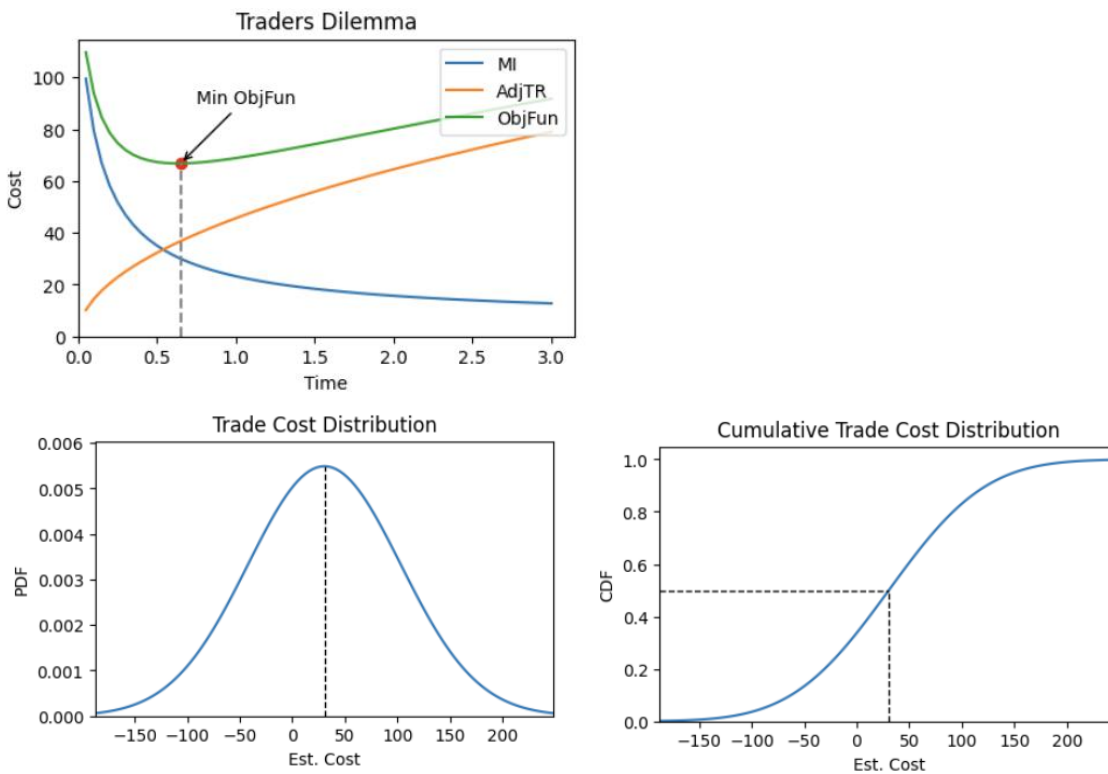
Questions:

Q1) Traders Dilemma

Solve the Traders Dilemma and determine the optimal POV rate for the following order:
Symbol=RLK, Side=Buy, Size=0.10, Volatility=0.25, Lambda=0.5.

- What are the TCA cost components: Optimal POV, MI, TR, Trade Time.
 - Optimal POV = 13.62%, MI=30.46, TR=72.68, POV=13.62%; TradeTime=0.6339
- Construct Traders Dilemma Graph
- Construct Trade Cost Distribution Graph (PDF)
- Construct Cumulative Trade Cost Distribution Graph (CDF)

Graphs:

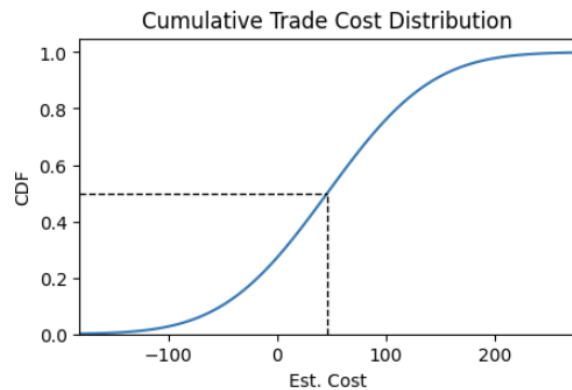
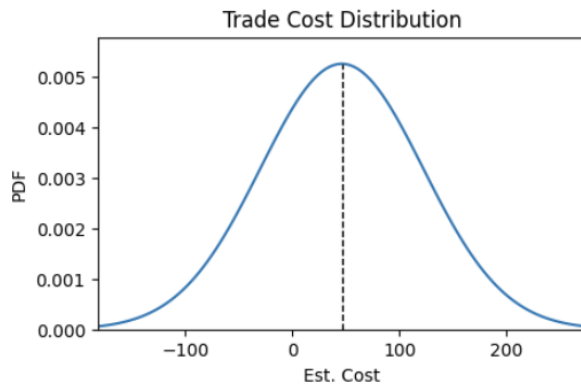
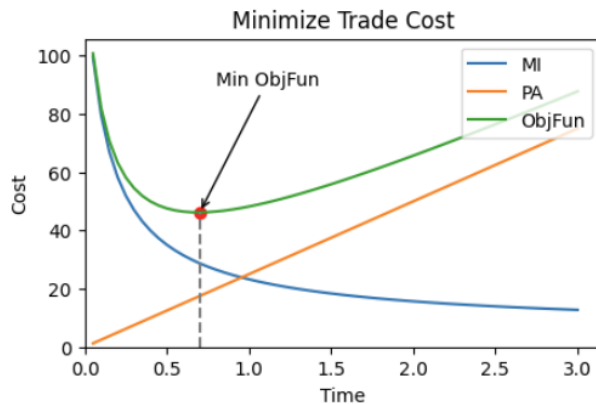


Q2) Minimize Cost

Solve the Minimize Cost trade strategy optimization problem for the following order: Symbol=RLK, Side=Buy, Size=0.10, Volatility=0.25, AlphaBp=100.

- What are the TCA cost components: Optimal POV, MI, TR, PA, and Trade Time.
 - Optimal POV=12.64%, MI=28.93; PA=17.27; TR=75.88; TradeTime=0.6909
- Construct Minimize Trade Cost Graph as function of Time
- Construct Trade Cost Distribution Graph (PDF)
- Construct Cumulative Trade Cost Distribution Graph (CDF)

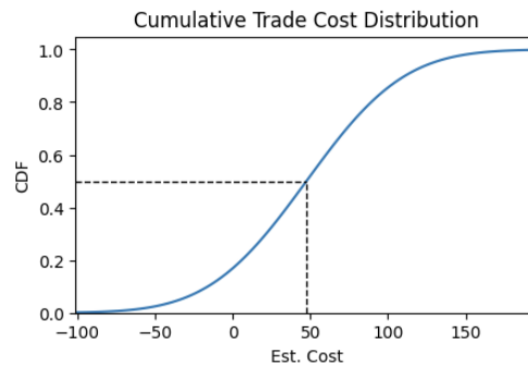
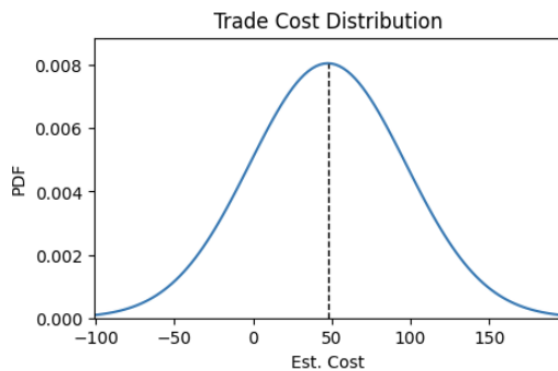
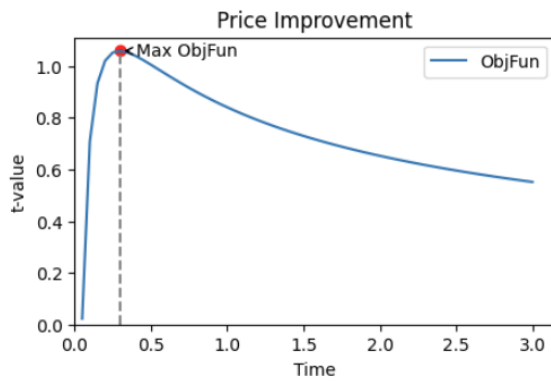
Graphs:



Q3) Price Improvement

Solve the Price Improvement trade strategy optimization problem for the following order:
Symbol=RLK, Side=Buy, Size=0.10, Volatility=0.25, Bid=100

- What are the TCA cost components: Optimal POV, MI, TR, PA, and Trade Time.
 - Optimal POV=25.35, MI=47.53; TR=49.52; POV=0.2535; TradeTime=0.2943
- Construct Minimize Trade Cost Graph as function of Time
- Construct Trade Cost Distribution Graph (PDF)
- Construct Cumulative Trade Cost Distribution Graph (CDF)
- **Graphs:**



Q4) Efficient Trading Frontier

Construct the Efficient Trading Frontier for the following Order:

Symbol='RLK', Side=Buy, Shares=150,000, ADV=2,000,000, Volatility=30%, AlphaBp=0.

Y-Axis = Market Impact (MI)

X-Axis = Timing Risk (TR)

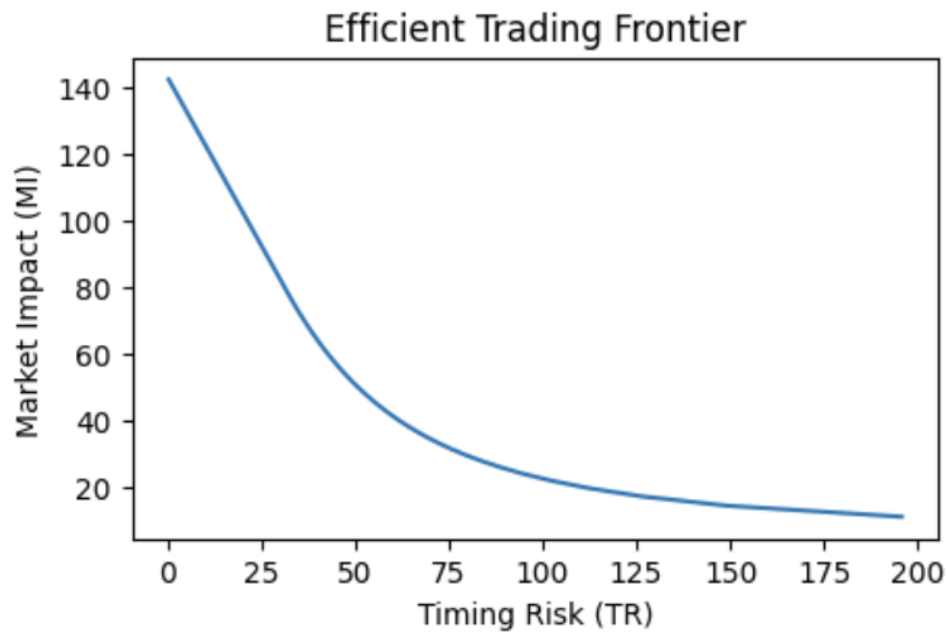
Hint:

Solve the Traders Dilemma Optimization for different values of Lambda.

Minimize $MI + \text{Lambda} * TR$

Use Lambda=0.05, 0.10, 0.15,, 2.95, 3.00.

Graph:



Q5) Market Impact Constraint

What is the maximum number of shares that can be executed over one full day (e.g., Trade Time = 1 day) with a maximum market impact of MI=50.

The order characteristics are:

Symbol='RLK', Side=Buy, ADV=2,000,000, Volatility=25%, AlphaBp=0, Trade Time = 1 day.

- Shares = 449,119; MI = 49.9999

Graph:

